Grassroots Ride-Hailing

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Abstract. Location-based services match a nearby requester and provider: a ride, a room, a delivery. The platforms that provide them hold every provider's data, match every request and charge a fee, and are a single point of control and of failure.
Here we present *grassroots ride-hailing*: a framework the participants run by themselves, with no central server and no coordinator, that keeps its properties as it grows from one local group to many distant ones. Removing the platform raises two problems, building the network and matching over it, and we address both. We specify matching as grassroots contracts, each party holding its own request or ride. The Grassroots Directed Overlay contract lets a person point to their friends and to the persons those point to; its location-based instance keeps each person pointing to a bounded number of the nearest of them, and we prove that a person and a person they point to are in one connected component of the graph of the friendships made. The Grassroots Ride Auction contract lets a rider call for bids from the persons they point to and the persons those point to, a driver bid a fare and a pickup delay, and the ride be awarded by the terms; its second-score instance scores a bid by the rider's willingness to pay less the fare, awards the ride to the highest score at the fare that leaves the winner the margin over the runner-up, and settles a late pickup by a refund out of the locked fare.

Keywords: Grassroots ride-hailing · Location-based services · Directed overlay · Grassroots contracts · Second-score auction · Grassroots platforms

1 Introduction

Location-based services match a nearby requester and provider: a ride, a room, a delivery. The platforms that provide them, Uber, Airbnb and Wolt among them, share one structure: a central platform that holds every provider's data, matches every request and charges a fee. The platform is a single point of control and of failure: it sets the prices, owns the data, and can exclude anyone.

This work proposes a **grassroots framework** [9] that the participants run by themselves, with no central server and no coordinator, and that keeps its properties as it grows from one local group to many distant ones. Removing the platform raises two problems, building the network and matching over it, and we address them both.

An agent is a person together with the smartphone that runs the person's code: a driver is a person and the phone that runs the driver code, and likewise a rider. The framework is specified by grassroots contracts [4], social contracts that run, written as clauses their parties undertake towards one another, from

which the app is compiled down the abstraction cascade [5] to Grassroots Logic Programs [10] and to a smartphone.

Section 2 builds the network, the Grassroots Directed Overlay over the friendships of the grassroots social graph [9,11]: each person points to their d nearest among their friends and the persons those point to, so that every pointing lies within a component of the friendship graph. Section 3 specifies the ride as the Grassroots Ride Auction contract, by which a rider calls for bids over the overlay, drivers bid a fare and a pickup delay, and the ride is awarded and settled by the terms. Section 4 shows the app, Section 5 reports a simulation, Section 6 reviews related work and Section 7 concludes.

2 Grassroots Directed Overlay

A rider needs a driver nearby, and with no platform that holds every driver's position, each agent has to find them. Agents are friends on the grassroots social graph [9,11], and a friend is not necessarily nearby, nor a nearby driver a friend. The Grassroots Directed Overlay builds, over the friendships, a second graph: each person points to friends and to the persons those point to.

GDO: Grassroots Directed Overlay

GDO extends GSG [4].
Alice, Bob and Carol are persons.

1. ■ *Pointing to friend.* Alice, who does not point to her friend Bob, may point to Bob.
2. ■ *Transitive pointing.* Alice, who knows Carol and points to Bob who points to Carol, may point to Carol.
3. ■ *Ceasing to point.* Alice, who points to Bob, may cease pointing to Bob.

Pragmas

Pointing: the persons Alice points to.
Pointing to friend: `Alice [Point]`
Ceasing to point: `Alice [Unpoint]`

GDO-LB: Location-Based GDO

GDO-LB restricts GDO.
Alice, Bob and Carol are persons. A person has a location. $d > 0$ is an integer and $e > 0$.

1. ■ *Pointing to friend.* GDO:1, only if either Alice points to less than d persons and Bob is closest to Alice among all of Alice's unpointed friends, or Bob is closer to Alice by more than e than some person pointed to by Alice.
2. ■ *Transitive pointing.* GDO:2, only if Carol is closer to Alice by more than e than some person pointed to by Alice.
3. ■ *Ceasing to point.* GDO:3, only if Alice points to more than d persons, with Bob furthest from Alice among them.

Definition 2.1 (Cumulative friendship graph). *The* ***cumulative friendship graph*** *of a run is the undirected graph whose vertices are the persons and whose edges are the pairs of persons that are friends at some configuration of the run.*

A pointing stays when the friendship it was made on ends (GSG:2), so the proposition is on the friendships that have ever held.

Proposition 2.2. *In every run of GDO, if Alice points to Bob, then Alice and Bob are in one connected component of the cumulative friendship graph of the run.*

Proof. By induction on the length of the run. At its start no person points to anyone. An act of GDO:3 removes a pointing and keeps the claim, and an act of GSG adds or removes no pointing; an edge of the cumulative friendship graph, once present, stays. An act of GDO:1 makes Alice point to Bob, who is Alice's friend, so the two are adjacent in the graph. An act of GDO:2 makes Alice point to Carol, where Alice points to Bob and Bob points to Carol: by the claim before the act Alice and Bob are in one component, and so are Bob and Carol; hence Alice and Carol are.

The overlay is the directed graph of the pointings. By Proposition 2.2 every pointing edge lies within a component of the cumulative friendship graph, so two persons that have never been joined by a chain of friendships are never joined by a chain of pointings. GDO-LB makes each person point to their d nearest among the persons they may point to, so the pointing edges are short.

3 The Ride

A ride is a sequence of functions: a passenger requests it, a driver takes it, and one of them ends it. We specify them as the clauses of a grassroots contract, in which the request is a call for bids, the take is a bid and an award, and the end is a pickup, an arrival or a cancellation that settles the fare.

[Ohad: Which driver takes which request is left open here][Udi: Removed.]

The ride auction. Over the overlay of Section 2, the three functions are the clauses of the Grassroots Ride Auction. The request is a *Call*: Alice calls for bids from the agents they point to and the agents those point to, stating the pickup point, the destination, the bidding period and the terms. The take is a *Bid* and an *Award*: each driver who received the call bids a fare and a pickup delay, and when the bidding period ends the ride is awarded to the bidder the terms select, at the fare the terms set, locking the rider's fare and the driver's deposit. The end is a *Pickup* followed by an *Arriving*, or a *Cancelling* when no pickup comes in time; either releases the fare and the deposit as the terms say.

GRA: Grassroots Ride Auction

GRA extends GDO.
Alice is a rider, Bob a driver and Carol a person. A person can be either a rider or a driver but not both. A person has a location.

1. ■ *Call.* Alice, who has no open call and no ride, may call for bids on a ride from the persons Alice points to and the persons they point to, stating the pickup point, the destination, the bidding period and the terms.
2. ■ *Bid.* Bob, who received the call, may bid a fare and a pickup delay before the bidding period ends; Alice may not open a bid before then.
3. ■ *Award.* When the bidding period ends, Alice awards the ride to the bidder the terms select among the bidders who have no ride, at the fare the terms set, and Alice's fare and the deposit the terms set for Bob are then locked; if the terms select no bidder, the call is closed.
4. ■ *Pickup.* The pickup is recorded when Alice and Bob are both at the pickup point.
5. ■ *Arriving.* When Alice and Bob are both at the destination, the fare and the deposit are released to Alice and to Bob as the terms say.
6. ■ *Cancelling.* If no pickup is recorded within the maximum pickup delay, Alice may cancel, and the fare and the deposit are released to Alice and to Bob as the terms say.
7. ■ *Withdrawing.* Alice may withdraw her open call before the bidding period ends.

GRA-SS: Grassroots Ride Auction, Second-Score Restriction

GRA-SS restricts GRA.
Alice is a rider, Bob a driver and Carol a person. The terms of a call state a maximum pickup delay $T \geq 0$, a willingness to pay $W(\tau) = A - \lambda\tau$ with $A, \lambda \geq 0$, and a deposit $D \geq 0$. A pickup delay is measured from the award, at time t_0. A bid (p, τ) is the fare $p \geq 0$ the bidder asks and the pickup delay $\tau \geq 0$ the bidder promises. Its score is $W(\tau) - p$, and it is admissible iff $\tau \leq T$ and its score is at least 0.
1. ■ *Transitive pointing.* GDO:2, where Carol is a driver.
2. ■ *Award.* GRA:3, where the terms select the bidder whose admissible bid has the highest score, ties broken as the terms say, or no bidder if no bid is admissible. They set the fare $P = W(\tau) - s_2$, where τ is that bid's pickup delay and s_2 is the highest score among the other admissible bids, or 0 if there is none, and they set the deposit D.
3. ■ *Arriving.* GRA:5, where the refund $\min\{P, \lambda\delta\}$ is released to Alice, and the rest of the fare and the deposit to Bob, with $\delta = \max\{0, t_{\text{pickup}} - t_0 - \tau\}$.
4. ■ *Cancelling.* GRA:6, where the fare and the deposit are released to Alice if Alice was at the pickup point from t_0 to $t_0 + T$, and otherwise the fare to Alice and the deposit to Bob.

Riders are at the rim of the overlay: a rider is pointed to only by their friends, by Pointing to friend, since transitive pointing is only to a driver.

The terms can be extended with a value of each bundle of driver and vehicle attributes the rider prefers, which enters the score of a bid and the fare.

4 The App

The contracts GDO and GRA are checked syntactically grassroots and compiled down the abstraction cascade [4,5] to the mini-apps of the grassroots app; their screens follow.

5 Simulation

Riders and drivers moving over a city, simulated in sGLP [12]: the pointings the Grassroots Directed Overlay keeps as agents move, and the fares, pickup delays and cancellations of the rides the auction awards, as the drivers and riders grow.

6 Related Work

Ride-hailing platforms and cooperatives. Uber and Lyft match every request on a central platform and take a fee [7,8]; a ride-sourcing market with a third-party integrator is studied in [15]. Platform cooperatives [1,14] such as The Drivers Cooperative [13] and Eva [3] are owned by their drivers but still run a central platform. Grassroots ride-hailing has no platform: the network and the matching are run by the participants.

Grassroots platforms and contracts. Grassroots platforms are due to [9], with social graphs, currencies and federations as atomic transactions [11]; a grassroots contract [4] is a social contract that runs, compiled through volition-guarded transactions [6] and the abstraction cascade [5] to Grassroots Logic Programs [10]; a lost key or device is recovered by [2], and people are simulated by stochastic agents in sGLP [12].

Scoring auctions.

7 Conclusion

Riders and drivers run ride-hailing by themselves, under two grassroots contracts and with no platform: the Grassroots Directed Overlay builds the network, each person pointing to the nearest of their friends and of the persons those point to, and the Grassroots Ride Auction matches over it, a rider calling for bids from the agents they point to, drivers bidding a fare and a pickup delay, and the ride awarded and settled by the terms. The contracts are compiled to a working app.

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